

PREDICATES AND NESTED QUANTIFIER

- ❖ Predicates and Quantifiers
- ❖ Translating English Sentences into Logical Expressions

Learning Objective

- ❖ To understand how to establish the correctness of computer programs.
- ❖ To understand how to translate the English sentence into simple and useful logical expressions.

Predicates

❖ Statements involving variables , such as

" $x > 3$ ", " $x = y + 3$ ", " $x + y = z$ "

and

" computer x is under attack by an intruder",

and

" computer x is functioning properly "

“ x is greater than 3 ”

the subject of the statement

the predicate

❖ Denote the statement “ x is greater than 3 ” by $P(x)$, where P denotes the predicate “ is greater than 3 ” and x is the variable.

Example 1 Let $P(x)$ denote the statement “ $x > 3$ ”. What are the truth values of $P(4)$ and $P(2)$?

Solution $P(4)$ means the statement “ $4 > 3$ ”, therefore $P(4)$ is **true**.

$P(2)$ means the statement “ $2 > 3$ ”, therefore $P(2)$ is **false**.

Example 2 Let $Q(x, y)$ denote the statement “ $x = y + 3$.” What are the truth values of the propositions $Q(1, 2)$ and $Q(3, 0)$?

Solution

For the statement “ $1 = 2 + 3$,” the truth value of $Q(1, 2)$ is **false**.

For the statement “ $3 = 0 + 3$,” the truth value of $Q(3, 0)$ is **true**.

Example 3 Let $A(c, n)$ denote the statement computer “c” is connected to network n where c is a variable representing a computer and n is a variable representing a network. Suppose that the computer **MATHS1** is connected to network **CAMPUS2** but not to network **CAMPUS1**. What are the values of $A(\text{MATHS1}, \text{CAMPUS1})$ and $A(\text{MATHS1}, \text{CAMPUS2})$?

Solution $A(\text{MATHS1}, \text{CAMPUS1})$ is false because **MATHS1** is not connected to the CAMPUS1 network.

$A(\text{MATHS1}, \text{CAMPUS2})$ is true because **MATHS1** is connected to the CAMPUS2 network.

Preconditions and Postconditions

- ❖ Predicates are also used to establish the correctness of computer programs, that is, to show that computer programs always produce the desired output when given valid input.
- ❖ The statements that describe valid input are known as **preconditions**.
- ❖ The conditions that the output should satisfy when the program has run are known as **postconditions**.

Preconditions and Postconditions (Example)

- ❖ Consider the following program, designed to interchange the values of two variables x and y .

temp := x

$x = a, \text{temp} = a, y = b$

$x := y$

$x = b, \text{temp} = a, y = b$

$y := \text{temp}$

$x = b, \text{temp} = a, y = a$

- ❖ For the precondition, the predicate $P(x, y)$ is the statement “ $x = a$ and $y = b$ ” where a and b are the values of x and y before we run the program.
- ❖ For the postcondition, the predicate $Q(x, y)$ is the statement “ $x = b$ and $y = a$ ”

Translating English Sentences into Logical Expression

EXAMPLE Express the statement “ If a person is female and is a parent, then this person is someone’s mother” as a logical expression involving predicates, quantifiers with a domain consisting of all people, and logical connectives.

Solution The statement “ If a person is female and is a parent, then this person is someone’s mother” can be expressed as “ **For every person x , if person x is female and person x is parent, then there exists a person y** such that person x is the mother of person y .”

Translating English Sentences into Logical Expression

$F(x)$ to represent “ x is female,”

$P(x)$ to represent “ x is a parent,” and

$M(x, y)$ to represent “ x is the mother of y .”

The original statement can be expressed as $\forall x \left((F(x) \wedge P(x)) \rightarrow \exists y M(x, y) \right)$.

We obtain the logically equivalent expression $\forall x \exists y \left((F(x) \wedge P(x)) \rightarrow M(x, y) \right)$.

Summary

- ❖ Predicates (**Preconditions and Postconditions**)
- ❖ How to Translate the English Sentences into Logical Expressions

PREDICATES AND NESTED QUANTIFIERS



Let $P(x, y, z)$ be the statement “ $x + y = z$.” What is the truth value of $P(1, 2, 3)$?

Answer: **TRUE**



Next Lecture



Predicates and Nested Quantifiers(Cont'd)